

Facial Emotion Recognition

A Deep Learning Approach

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Problem Statement

- ~55% of emotional meaning is conveyed visually through facial cues
- Emotion recognition is critical but challenging at scale
- Affective Computing enables machines to interpret emotions
- Facial Emotion Recognition is a key component of human-centered AI

Goal: Develop a deep learning model that can accurately classify facial emotions from images.

Dataset Overview

- Consists of labeled images of faces
- Divided into train, validation, and test sets
- Grayscale images (48 × 48 resolution)
- 4 emotion classes:



Happy



Sad



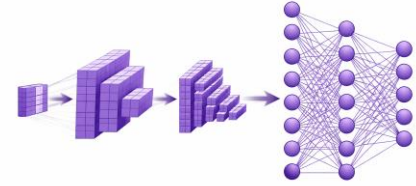
Neutral



Surprise

Modeling Approach

- Built and refined **custom CNN architectures** for feature learning



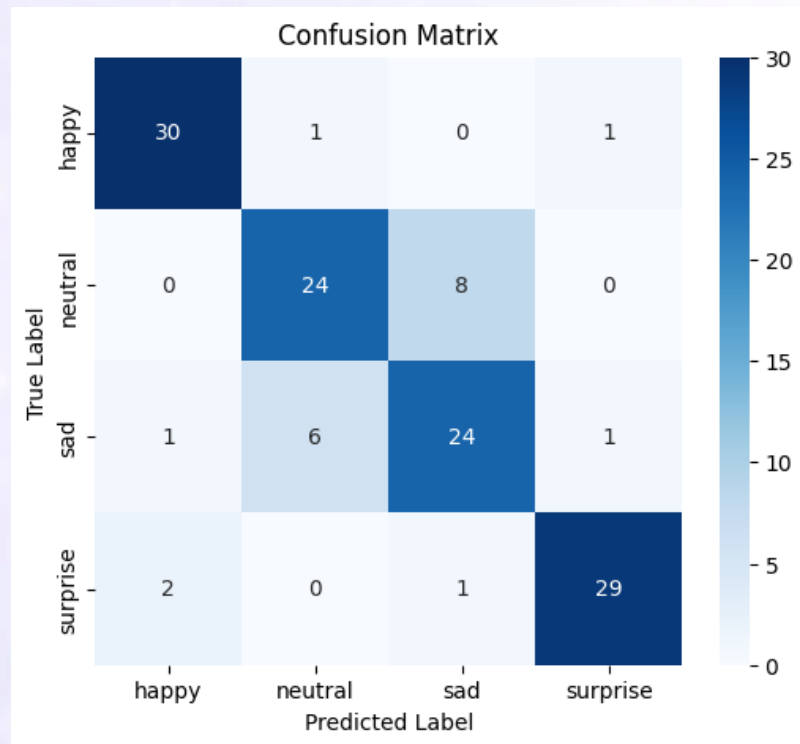
- Benchmarked against **transfer learning models**
(VGG16, ResNet, EfficientNet)



- Conducted **multi-metric evaluation** and class-level error analysis



Model Performance



Model	Test Accuracy	Test Loss	Precision	Recall	F1-score
CNN model 1	71.9%	0.665	0.73	0.72	0.72
VGG16	72.7%	0.728	0.75	0.73	0.73
ResNet	75.0%	0.828	0.76	0.75	0.75
EfficientNet	75.8%	0.808	0.78	0.76	0.76
Complex CNN	81.3%	0.81	0.81	0.81	0.81
CNN Model 2	83.6%	0.454	0.84	0.84	0.84

- Custom CNN achieved
 - ~84% test accuracy
 - Lowest loss
- Transfer learning showed weaker performance

Key Insights



Custom CNNs
outperformed
transfer learning
models



Transfer learning
limitations in FER



Emotion separability
varies across classes



Model complexity
does not guarantee
better performance



Model selection
depends on
generalization, not
just accuracy



Subtle emotions are
harder to distinguish

Recommendation



Proceed with CNN Model 2 as the preferred model

- Production-ready baseline



Use the model with caution in applications involving subtle emotions



Position the model as a decision-support tool rather than a standalone decision-maker



Invest further if the target use case requires higher reliability on subtle emotions

- Data quality and representation

Real-World Applications



Customer experience analytics



Education technology



Human-computer interaction (HCI)



Facial Emotion Recognition



Mental health monitoring tools



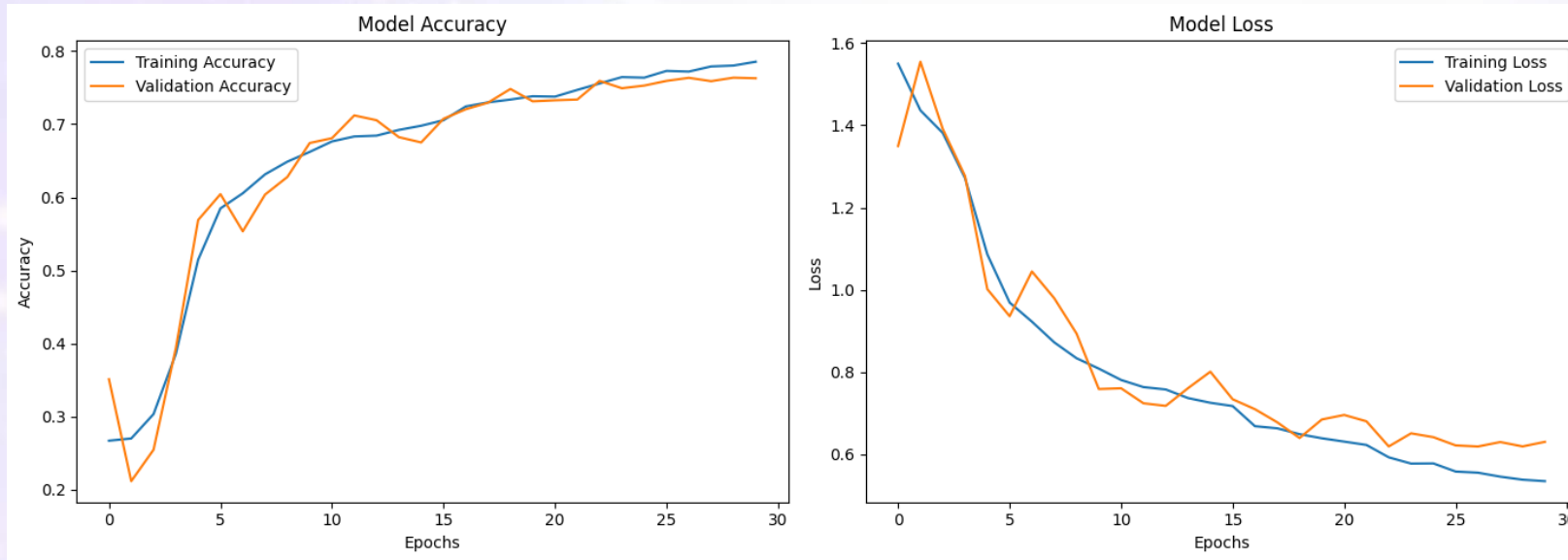
Driver fatigue and emotion detection systems

The background of the slide features a woman's profile on the right side, with a white geometric overlay resembling a facial recognition or neural network pattern on her face. The left side of the background is filled with various semi-transparent icons and graphics: a grid of ten small portrait photos, a line graph with multiple peaks, a bar chart, a smiley face icon, a sad face icon, and a brain-like structure with glowing nodes. The overall color palette is light and airy, with soft purples and blues.

**Thank you.
Questions?**

Appendix

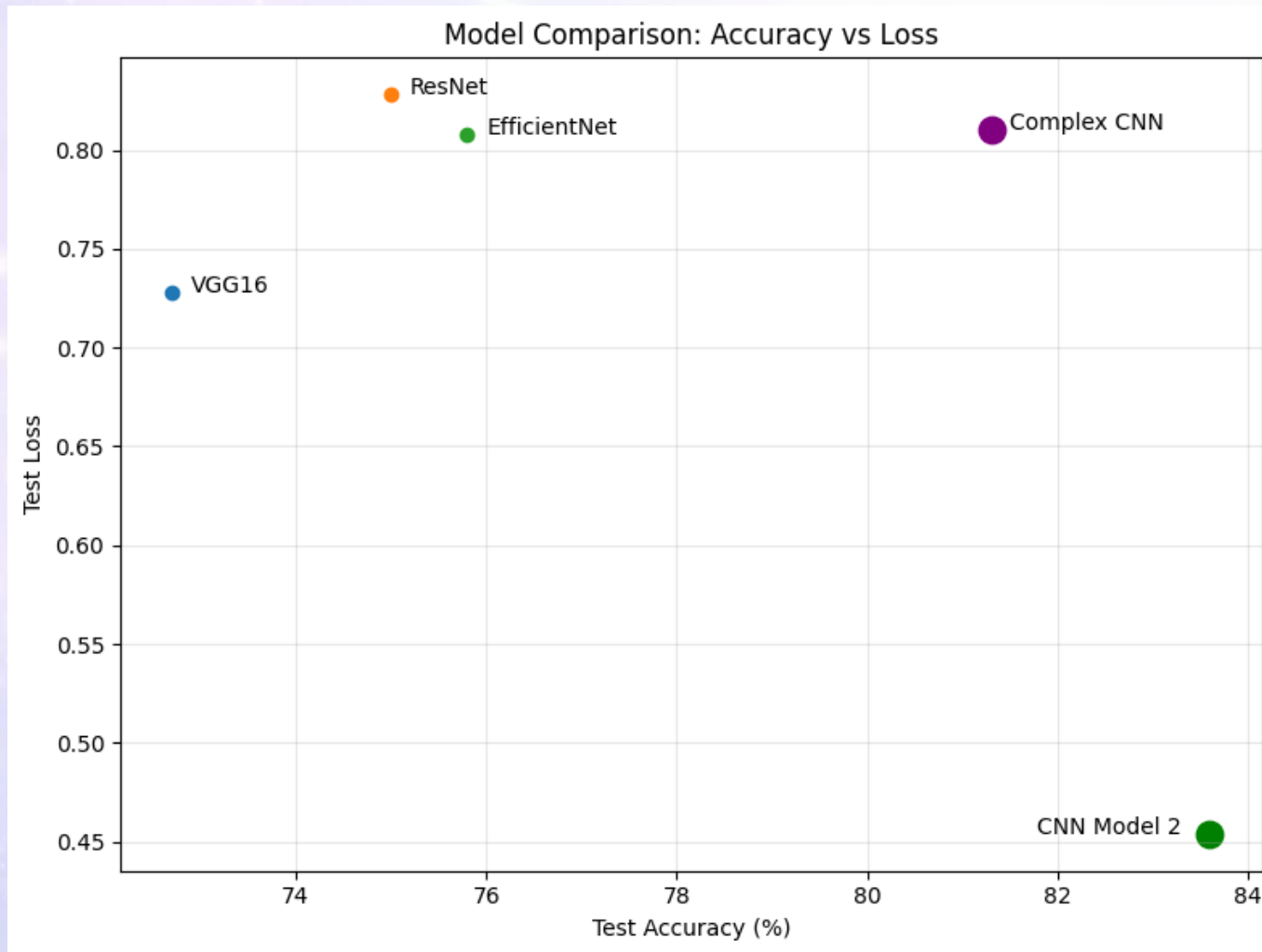
Training Stability and Generalization Performance



- Stable learning
- Good generalization
- No signs of strong overfitting

Appendix

Model Comparison



- Optimal region achieved by CNN Model 2
- Clear performance gap between model families